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The influence of flooding imagery and party cues on perceived threat, collective efficacy, and intentions for political action to address climate change.

Abstract

This study investigates how episodic and thematic flooding imagery and political party cues in climate change news stories influence risk perceptions, collective efficacy perceptions, and intended political action. We found that imagery and party cues had significant total effects on perceived flooding threat but not on perceived climate change threat. Specifically, episodic imagery (but not thematic imagery) increased perceived flooding threat, and party cues lowered perceived threat. Perceived threat was positively associated with collective efficacy, which, in turn, was positively associated with intended political action. Both imagery and party cues had significant indirect effects on perceived efficacy and intended action.

Keywords. Climate Change Communication, Imagery, Party Cues, Risk Perceptions, Collective Efficacy, Political Action

The influence of flooding imagery and party cues on perceived threat, collective efficacy, and intentions for political action to address climate change.

The overwhelming scientific consensus is that climate change is real, human caused, and already producing significant negative impacts throughout the world (IPCC, 2014). Despite the urgent need for action to address climate change, the issue remains highly polarized in the United States (Dunlap et al., 2016). Even among those concerned about climate change, most have not taken political action on the issue (Roser-Renouf et al., 2014). As news media are one of the primary ways that the public learns about climate change (Feldman et al., 2012), a key question for understanding public motivation to act is how aspects of climate change news coverage may influence public perceptions and engagement.

This study examines the effects of two increasingly central characteristics of climate change news: imagery of extreme weather and political party cues. Imagery, as used in this study, refers to visuals in news stories used to depict phenomena, and includes content such as vivid photographs of extreme weather and maps depicting the potential risks of extreme weather. Party cues refer to information about where members of a political party stand on an issue (Bullock, 2020), such as the degree to which politicians or lay partisans believe that climate change is making extreme weather events more likely to occur.

The current scientific consensus is that climate change contributes to extreme weather (National Academies of Sciences, Engineering, and Medicine, 2016). Extreme weather events such as floods provide opportunities for news organizations to incorporate climate change in their coverage and discuss the link between the two phenomena (Painter et al., 2021; Weiner et al., 2021); such news coverage often includes strong imagery depicting the extreme weather

events (Nerlich & Jaspal, 2014). Images have a potentially powerful yet relatively understudied influence on public engagement with climate change (O'Neill, 2013, 2020; O'Neill & Smith, 2014). Thus, the present study adds to the still nascent literature on how climate change imagery in news stories may affect public perceptions and inclinations to take action on the issue by comparing the effects of episodic (e.g. a photograph of a flooded house) and thematic (e.g. a map of flooding risk based on statistical data) images of flooding. Given the politically polarizing nature of climate change, we also examine how political ideology may moderate effects of imagery.

At the same time, recent research has demonstrated that news coverage about climate change has become increasingly politicized in the U.S. (Chinn et al., 2020), meaning that it highlights political divisions between Republicans and Democrats and often includes explicit cues about where political parties stand on the issue (Merkley & Stecula, 2018). Alongside politicized news coverage, the U.S. public is strongly divided in their perceptions of climate change's contribution to extreme weather (Leppert, 2022). It is thus important to study the impact of party cues that highlight this polarization in climate change news stories, and how individuals with varying political orientations may respond to such cues, given past research demonstrating the effects of party cues on public perceptions (Bullock, 2020).

The present study offers a novel examination of how imagery and party cues may interact in a climate change context. While some work (Bolsen et al., 2019a) has examined how the inclusion of imagery may counteract skeptical climate change frames, we are not aware of previous work that has examined how climate imagery may moderate party cues. This is an important research question, given that current literature suggests that imagery may attenuate

political polarization (Bolsen et al., 2019a) while party cues may amplify polarization (Levendusky, 2010) on issues where strong party divides exist such as global climate change. This study helps address this research gap by investigating how imagery and party cues may influence risk perceptions around climate change and flooding, as well as the downstream outcomes of perceived collective efficacy and intentions for political action.

Risk perceptions have the potential to impact perceptions of collective efficacy, which is a belief that a group can collectively take action towards achieving a goal (Choi & Hart, 2021). A still nascent area of research suggests that risk perceptions of climate change may be positively associated with perceived efficacy through motivated control processes (Hornsey et al., 2015). According to this theoretical framework, individuals may try to counteract the negative feelings associated with a perceived threat by convincing themselves that they have the efficacy to manage the threat. While some previous research has suggested that impact imagery may inhibit feelings of efficacy (O'Neill & Nicholson-Cole, 2009), findings from motivated control literature suggest that the opposite may occur - impact imagery may actually increase perceptions of efficacy through the mediator of risk perceptions. Additional research is needed to better understand these processes.

Finally, there are conflicting findings regarding how collective efficacy is associated with intentions to engage in political actions to address climate change. While some research has found that individuals who feel a greater sense of collective efficacy are more likely to take political action (van Zomeren et al., 2008), other studies have found null results (Doherty & Webler, 2016). Given these conflicting findings, more research is needed to better understand how these constructs are associated with each other.

In the present study we utilize a moderated-mediation model to investigate how different types of party cues and impact imagery may impact risk perceptions of both flooding and climate change, which may, in turn, affect downstream variables of collective efficacy and intentions to engage in political behavior. This approach affords the opportunity to examine several research areas that have conflicting findings and is the first study we are aware of to investigate these message factors and theoretical constructs in combination with each other.

Literature Review

Imagery

While climate change imagery is widely seen as a critical aspect of climate news coverage (O'Neill, 2013, 2020; O'Neill & Smith, 2014), there remains a dearth of research examining the effects of climate imagery on public perceptions (O'Neill, 2017; Wang et al., 2018). Imagery is a powerful communication tool that operates through different communicative pathways than text (O'Neill, 2017). Images can be particularly effective in generating emotional responses from viewers and facilitating information recall (O'Neill & Smith, 2014). There is a need for additional research to understand how climate imagery may affect the public, in part because the limited research to date has shown mixed results (O'Neill, 2017). For example, research using Q-sort methodology suggests that impact imagery, such as an overhead view of a flooded area, may increase the feeling that climate change is important while also causing message recipients to feel overwhelmed (Leviston et al., 2014; O'Neill, 2013; O'Neill & Nicholson-Cole, 2009). Experimental approaches looking at similar types of imagery, however, have failed to replicate these results, instead finding that impact imagery has limited

effects on perceived issue importance and efficacy (Feldman & Hart, 2018; Hart & Feldman, 2016).

An important aspect of imagery that could affect public perceptions is whether the image provides *episodic* information by focusing on a vivid exemplar or *thematic* information that offers broader context about the issue at hand (Iyengar, 1990; Slovic, 2007). For example, when covering climate change, an episodic image might be a photograph of a flooded house, whereas a thematic image might be a map showing the risk of flooding across the United States. Some previous research suggests that episodic imagery may be particularly powerful in invoking concern and promoting action to address an issue (Rickard et al., 2017; Slovic, 2007; Small et al., 2007). Slovic (2007) finds that showing an image of a single victim, such as a child at risk of starvation, typically evokes more compassion, concern, and willingness to donate than providing more thematic information about how millions are at risk of starvation. When looking at the influence of imagery on perceptions of hurricanes, Rickard et al. (2017) found that a photo of a flooded house, compared to a risk map or no image, led to higher levels of risk perceptions and evacuation behavioral intentions. Exemplification theory (Zillmann, 1999) also posits that vivid exemplars, such as those offered by strong imagery (Zillmann, 2006), can outweigh information describing the issue as a whole, such as base rate statistics.

Other research, however, has found that providing thematic information about systemic risk can also be an effective messaging strategy (Bolsen et al., 2019a; Hart, 2011; Iyengar, 1990). For example, Iyengar (1990) found that thematic, rather than episodic, depictions of poverty in the U.S. led to greater support for government programs to address poverty. When individuals were exposed to individual exemplars of impoverished individuals, they were more

likely to ascribe responsibility for being poor to the individuals themselves, whereas when participants were provided thematic information about poverty throughout the U.S., they were more likely to attribute responsibility to society at large and tended to be more supportive of government programs to address poverty (Iyengar, 1990). Thus, thematic representations of an issue can lead to greater support for action through the view that society at large is causing the issue and that collective action through policies or other mechanisms can address it. In the context of climate change, Bolsen et al. (2019a) found that the inclusion of thematic maps that depicted future projections of severe weather events across the U.S. was moderately successful in reducing politicization around climate change. For example, the inclusion of a map showing how climate change is likely to increase the frequency of extreme heat events in the U.S. appeared to counteract the influence of a skeptical climate change frame when participants were asked to assess the impacts of climate change on the weather.

Given the conflicting results in previous literature, the present study examines how news stories stating that climate change is increasing flooding risks that include vivid, episodic imagery, such as a flooded house, compare to those that provide thematic imagery, such as a map of how climate change is increasing flooding risks across the U.S., in their effects on the perceived threat, efficacy, and willingness to take action to address climate change. We investigate how these different types of imagery may increase two types of threat perceptions: (a) the threat of flooding and (b) the threat of climate change.

While the perceived threat of flooding and climate change are likely to be correlated, the perceived risk of flooding may be more susceptible to message influences given the highly polarized nature of climate change beliefs (Funk & Hefferon, 2019; McCright & Dunlap, 2011).

As a result of the polluted information environment around climate change (Kahan, 2012; McCright et al., 2016) and the intensifying connection between climate change and political identity (Nisbet, 2009), message strategies tend to have very small effects on climate change beliefs (Rode et al., 2021). While perceived flooding risk may be more susceptible to messaging than perceived climate risks, overall, we expect that the inclusion of imagery will increase the perceived threat of both flooding and climate change. Formally stated, we investigate the following hypotheses and research question:

H1a: A climate change news story that includes flood imagery will increase the perceived threat of *flooding* relative to a news story without imagery.

H1b: A climate change news story that includes flood imagery will increase the perceived threat of *climate change* relative to a news story without imagery.

RQ1: Will vivid, episodic impact imagery differentially influence the perceived threat of flooding and climate change compared to thematic imagery?

Party Cues

In addition to imagery, party cues (i.e., messages that signal where a political party stands on an issue like climate change) can change a person's views on that issue by making them more likely to adopt their party's stance (Bullock, 2020). "Polarized" cues that highlight stark partisan divisions on an issue (as opposed to more moderate divisions) are especially likely to influence citizens' attitudes (Levendusky, 2010). On the issue of climate change, Guber (2013) has demonstrated that differences in public opinion along party lines follow from the public's attention to and familiarity with cues from opinion elites such as politicians.

Prior research has also linked perceived partisan divisions regarding climate change – likely caused in part by exposure to party cues - to lower risk perceptions; the impression of conflict and confusion may cause people to detach from or disregard an issue (Linde, 2020). For example, using survey data from Sweden, Norway, New Zealand, and Australia, Linde (2020) not only found that people’s opinions follow the cues of their political party but also that people report less concern about climate change when they perceive more polarization among elites. A comparative analysis within the European Union also found that in countries where elites are united in support of environmental issues, the public is more likely to perceive climate change as a serious threat than in countries where party elites are divided (Sohlberg, 2017).

Given the increasing availability of party cues in news coverage of climate change (Chinn et al., 2020; Merkley & Stecula, 2018), it is important to understand how these cues affect audiences. Notably, news stories can highlight partisan divisions between elites (i.e., members of Congress) or between ordinary members of the public (i.e., lay partisans). While prior research has examined the effects of elite cues (Levendusky, 2010) and public cues (Levendusky & Malhotra, 2016), we are not aware of any research that has compared the two or that has done so in a climate change context. Thus, we examine how party cues in news stories influence the public’s perceptions of climate change and whether these effects vary based on whether the cues depict divisions among elites or general citizens. Formally stated, we investigate the following hypotheses and research question:

H2a: A climate change news story that includes party cues highlighting the political conflict around climate change will decrease the perceived threat of *flooding* relative to a news story without party cues.

H2b: A climate change news story that includes party cues highlighting the political conflict around climate change will decrease the perceived threat of *climate change* relative to a news story without party cues.

RQ2: Will elite party cues (Congressional representatives) differentially influence the perceived threat of flooding and climate change compared to party cues from lay partisans regarding the issue?

Interactions Between Conditions and Moderation by Political Ideology

In addition to examining the main effects of imagery and party cues, we are interested in how they may interact to influence public opinion. Current literature suggests that party cues and imagery may impact climate change polarization in different ways. While party cues may amplify polarization as partisans align their views with the divided political norm, recent research suggests that the inclusion of imagery depicting negative impacts of climate change may attenuate politically polarizing responses to climate change information (Bolsen et al., 2019a). We are unaware, however, of previous research that has examined how different combinations of party cues that highlight polarization and imagery may influence risk perceptions of climate change and extreme weather events, respectively. While the inclusion of conflicting party cues may lower climate risk perceptions (Linde, 2020), showing imagery of flooding that may result from increased extreme weather caused by climate change may counteract this effect. We formally hypothesize the following:

H3a: The presence of party cues and impact imagery in climate change news stories will interact to influence risk perceptions of *flooding*.

H3b: The presence of party cues and impact imagery in climate change news stories will interact to influence risk perceptions of *climate change*.

Political ideology may also interact with imagery and party cues in shaping public opinion. Climate change is one of the most politically polarizing issues in the U.S. today (Funk & Hefferon, 2019; Pew Research Center, 2020), which often leads to different responses from liberals and conservatives to climate change information (Dixon et al., 2017; Druckman & McGrath, 2019; Garrett, 2017; Hart & Nisbet, 2012; Nisbet, 2009). While some previous work has found that climate change beliefs may be more malleable for conservatives than liberals (Jenkins-Smith et al., 2020; Schuldt et al., 2011), there are also a number of studies in which climate change message impacts are not moderated by political ideology (Campbell-Arvai et al., 2017; Hart et al., 2022), as well as studies showing that liberals may be more responsive to experience with climate change impacts (Ogunbode et al., 2017). Given the conflicting results in previous literature, we ask as a research question:

RQ3: Will political ideology moderate the influence of party cues and impact imagery on public perceptions of climate change and flooding?

The Influence of Perceived Threat on Perceived Efficacy

Following the influence of climate change imagery and party cues on risk perceptions, we investigate how these risk perceptions may, in turn, influence perceived collective efficacy. As risk perceptions increase, individuals often engage in motivated control (Kay et al., 2009), in which greater threat perceptions are accompanied by a greater perceived ability to address the threat (Hornsey et al., 2015). Motivated control processes are driven by the fact that greater perceptions of threat can lead to negative feelings such as anxiety and discomfort; individuals

often compensate for these negative feelings by assuming a greater capability to address an issue (Kay et al., 2009). For example, if climate change imagery increases threat perceptions, individuals may boost their sense of efficacy as a way of coping with the unpleasant feelings induced by the increased perceived threat. Multiple studies examining climate change threat and efficacy perceptions have found that higher perceptions of threat correspond to increased efficacy perceptions (Heath & Gifford, 2006; Hornsey et al., 2015; Kellstedt et al., 2008; Milfont, 2012; Stoutenborough & Vedlitz, 2014). Thus, we expect that both flooding and climate change threat perceptions will have a positive association with perceived efficacy.

In the present study we focus on collective efficacy, which has received recent attention as a potential driver of political action (Landmann & Rohmann, 2020; van Zomeren et al., 2010). There is empirical support that the motivated control process affects both self and collective efficacy. However, the majority of motivated control studies have focused on perceptions of self efficacy (e.g., Heath & Gifford, 2006) with few examining collective efficacy (e.g., Hornsey et al., 2015). In the present study we focus on collective efficacy to add to the scant research utilizing this construct. Formally stated, we predict that:

H4a: The perceived risk of flooding will have a positive association with perceived collective efficacy to address climate change.

H4b: The perceived risk of climate change will have a positive association with perceived collective efficacy to address climate change.

Influence of Collective Efficacy on Intentions for Political Action

Collective efficacy is theorized to play an important role in driving behavior to address large-scale issues (Fritzsche et al., 2018). While any individual action alone is likely to have a

minimal impact on climate change, a large-scale collective shift in behavior can substantially shift expected outcomes. Thus, in the present study we focus on how collective efficacy may influence individuals' intentions to promote collective action through political behavior such as signing a petition, joining a protest, or writing to an elected representative to promote climate mitigation policies.

The link between collective efficacy and political action is still an under-researched area (Fritzsche et al., 2018), and studies examining this relationship have found mixed results regarding whether collective efficacy is a key determinant for political action. While some studies provide empirical support for the positive association between collective efficacy and political behaviors related to climate change (Chen, 2015; Landmann & Rohmann, 2020; van Zomeren et al., 2008), others suggest a less robust relationship (Choi & Hart, 2021; Doherty & Webler, 2016; Hamann & Reese, 2020). Thus, we pose the following research question:

RQ4: Will perceived collective efficacy have a positive association with intentions to take political action?

Given all of the hypothesized linkages, we investigate a mediation model such that imagery and party cues may influence the parallel mediators of the perceived risk of flooding and the perceived risk of climate change; these parallel mediators are predicted to have a positive association with the serial mediator of collective efficacy, which, in turn, is expected to have a positive association with the ultimate dependent variable of intended political action. Given the mixed findings regarding how liberals and conservatives may differentially respond to climate information, we also explore the research question of whether the effects of imagery and party cues on risk perceptions are moderated by political ideology. In addition to the

Hypotheses and Research Questions stated above and the proposed moderated-mediation model, we also examine whether climate change imagery and party cues may have any direct or total effects on perceived collective efficacy and intended political action, and whether such effects may be moderated by political ideology (see Figure 1).

Overall, our investigation draws from theories related to framing, exemplification, political party cue effects, and motivated control to provide a better understanding of how perceived threat, efficacy, and intentions for political action may be influenced by imagery and party cues in news stories. This study advances previous work focused on the effects of climate news visuals (Bolsen et al., 2019a; Rickard et al., 2017) by examining the interaction of visuals with party cues and political identity. We also extend the outcomes analyzed in previous work to include different types of perceived risk (i.e., flooding vs. climate change) as well as perceived efficacy and behavioral intentions, which allows us to test how imagery and party cues may directly and indirectly affect motivated control processes.

[Insert Figure 1 About Here]

Method

Experimental Design and Procedure

An online survey experiment was conducted using a 3 (*imagery*: no image, episodic image, thematic image) x 3 (*party cues*: no party cues, public party cues, elite party cues) plus control (no-message) between-subjects design. The stimulus, a news story constructed for this study, was embedded within an online Qualtrics survey.

Sample

An initial sample of 4,408 U.S. adults was recruited through Lucid, which pools participants from a variety of opt-in online survey sample suppliers; quotas were used to approximate national census distributions. Of these 4,408 participants, 1,153 failed a simple attention check question such as “please select ‘slightly agree.’” An additional 503 participants were removed from the analysis because they did not finish the survey or because their time to completion was below 1/3 or greater than three times the median completion time for their condition. This left 2,752 participants for the final analysis. The mean age was 47.14 ($SD = 17.24$), and 52.99% of the sample was female. The median level of education was “some college, no degree,” and the median household income before taxes was \$50,000 - \$74,000.

Procedure and Stimuli

After consenting to participate, participants were randomly assigned to see a news story corresponding to one of the nine treatment conditions or to a no-message control condition. Participants in the message conditions viewed a short news article constructed for this study that discussed how climate change is likely to increase the risk of flooding in the United States. In the public party cues conditions, the news story included information that a survey from the Pew Research Center found that a majority of Democrats are convinced that climate change is increasing the frequency and severity of climate change, while a majority of Republicans were skeptical. In the elite party cues conditions, the news story included similar information, but referred to Congressional Democrats and Republicans rather than the public at large. For participants in the episodic image conditions, the news story contained an image of a house that was flooded. For participants in the thematic image conditions, the news story contained a map showing the risk of flooding in different areas of the United States. Examples of the

stimulus are provided in the appendix (Figures A1 and A2). In order to improve external validity, stimulus sampling was utilized for the imagery conditions such that four episodic images¹ and two thematic images were randomly rotated between participants in the respective conditions (e.g. in the episodic image condition, participants were randomly assigned to see one of four possible episodic images). The episodic and thematic images were pre-tested for the study using a within-subject design (n=100), and when comparing between the episodic images (comparing the episodic images against themselves) and comparing between the thematic images, no significant differences were found for the variables of how concerned the image made the viewer about flooding, how personally relevant the image felt, how attention grabbing the image was, how easy or difficult it was to understand what was happening in the image, or how believable the image was. In addition, there were no differences between the respective stimulus sampled images (e.g. between the responses to the four episodic images) for any of the variables of interest used in the moderated-mediation model; given the similar responses to the stimulus sampling, the responses within each image condition (i.e., responses to the four episodic images and to the two thematic images, respectively) were collapsed together for the analysis.

After viewing the news story, participants answered questions to indicate their perceived threat of flooding, perceived threat of climate change, perceived collective efficacy to address the issue, and intentions to engage in political action related to climate change.

¹ We included more episodic images than thematic images for stimulus sampling because we found a greater variety in the types of episodic images of flooded houses in news stories. An important aspect of the episodic images was whether they were shot from ground level or an aerial shot from above; we wanted to make sure to account for this factor, so we used two ground level photographs and two aerial photographs for the stimulus sampling of episodic imagery.

Participants in the control condition did not view the news story, but answered identical questions as participants in the message conditions.

Measures

Perceived Threat of Flooding. The perceived threat of flooding was measured by asking participants “Overall, how much of a threat do you think floods pose?” on a scale that ranged from (1) no threat at all to (5) a very large threat, with (3) a moderate threat as the center point ($M = 3.65$, $SD = 1.08$).

Perceived Threat of Climate Change. The perceived threat of climate change was measured by asking participants “Overall, how much of a threat do you think climate change poses?” on the same threat scale as the perceived threat of flooding ($M = 3.58$, $SD = 1.31$).

Perceived Collective Efficacy. Perceived collective efficacy was measured by asking participants how much they agreed with each of the following three statements: (1) Americans can take joint action to address climate change, (2) Americans can come together to address climate change, and (3) Large numbers of Americans can collectively advocate for policies to address climate change. Each of these items was measured on a scale that ranged from (1) strongly disagree to (5) strongly agree, with (3) neither agree nor disagree as the center point. The three items were then averaged together for a single measure of collective efficacy (Cronbach’s $\alpha = .91$, $M = 3.85$, $SD = 1.08$).

Intentions to Engage in Political Action. Intentions to engage in political action were measured by asking participants how likely or unlikely they are to personally engage in the following activities in the next 12 months: (1) contact a government official to urge them to take action to reduce climate change, (2) participate in a rally or protest in support of action to

reduce climate change, (3) sign a petition in support of taking action to reduce climate change, (4) donate money to an organization working to reduce climate change, and (5) vote for a candidate for public office because of their support for action to reduce climate change. Each of these items was measured on a scale that ranged from (1) extremely unlikely to (5) extremely likely, with (3) neither likely nor unlikely as a center point. The five items were then averaged together for a single measure of intended political action (Cronbach's $\alpha = .92$, $M = 2.90$, $SD = 1.24$).

Political Ideology. Political ideology was measured by asking participants "Generally speaking, how would you best describe your political views?" on a scale that ranged from (1) very liberal to (7) very conservative, with (4) moderate or middle of the road as the center point ($M = 4.04$, $SD = 1.78$). Political ideology was treated as a continuous variable for all analyses.

Analytic Approach

We first used analysis of covariance (ANCOVA) to examine total effects and if there were two- or three-way interactions between imagery, party cues, and political ideology on any of the outcome variables. We then used PROCESS 4.2 (Hayes, 2022), which uses a path-based OLS regression analysis approach, to analyze the mediation models.

Results

First, ANCOVA tests were used to investigate main and interactive effects between imagery, party cues, and the political ideology of the participants on perceived threat of flooding, perceived threat of climate change, perceptions of collective efficacy, and intended political action (these examined the 3x3 interaction but excluded the no-message control variable).

Looking first to main effects of imagery messages (addressing H1a/b and RQ1), an ANCOVA (controlling for political ideology, age, and gender) found that imagery had a significant effect on the perceived threat of flooding ($F(2,2459) = 5.14, p = .006, \eta_p^2 = .004$), but did not have a significant effect on the perceived threat of climate change ($F(2,2458) = .03, p = .970, \eta_p^2 < .001$), perceived collective efficacy $F(2, 2461) = .31, p = .735, \eta_p^2 < .001$, or political action intentions ($F(2, 2461) = .06, p = .938, \eta_p^2 < .001$).

Sidak post hoc tests for the impact of imagery on the perceived threat of flooding found that episodic images ($EMM = 3.76, SE = .04$) significantly increased the perceived threat compared to thematic images ($EMM = 3.61, SE = .04, p = .007$) and no image messages ($EMM = 3.64, SE = .04, p = .041$). The thematic vs. no image comparison was not significant ($p = .918$).

Looking next to the main effects of party cues (addressing H2a/b and RQ2), the ANCOVA revealed that party cues had a significant effect on the perceived threat of flooding ($F(2,2459) = 4.80, p = .008, \eta_p^2 = .004$), but did not have a significant effect on the perceived threat of climate change ($F(2,2458) = 1.37, p = .256, \eta_p^2 = .001$), perceived collective efficacy $F(2, 2461) = .77, p = .454, \eta_p^2 = .001$, or political action intentions ($F(2, 2461) = .56, p = .571, \eta_p^2 < .001$).

Sidak post hoc tests for the impact of party cues on the perceived threat of flooding found that public cues ($EMM = 3.62, SE = .04$) significantly lowered perceived threat compared to the no party cue condition ($EMM = 3.76, SE = .04, p = .015$); elite cues ($EMM = 3.63, SE = .04$) significantly lowered perceived threat compared to no party cues ($p = .037$), and there was not a significant difference between elite and public cues ($p = .989$).

We also note that political ideology had a significant association with all variables (all $p < .001$), such that more liberal participants had a greater perceived threat of flooding, perceived threat of climate change, collective efficacy, and intentions for political action.

Looking to interactions (addressing H3a/H3b and RQ3), there were no significant 2- or 3-way interactions between imagery, party cues, and political ideology on any of the downstream variables (risk perceptions of flooding, risk perceptions of climate change, collective efficacy, or intentions for political action; all $ps > .05$).

Given that we did not find evidence of interactions between the message conditions or political ideology, we then used PROCESS to test an unmoderated mediation model in which imagery and party cues may influence political action intentions through the mediators of perceived threat of flooding, perceived threat of climate change, and collective efficacy (see Figure 1).

Multiple mediation models were run with different dummy coded variables as reference groups for comparing the indirect effects of the respective conditions. All analyses control for age, sex, and political ideology. When comparing results between message conditions, we exclude the no-message control variable so that both message factors can be controlled for without creating a singularity. See Table 1 for coefficients of the regression using the no-image and no-party cue message conditions as the reference groups.

[Insert Table 1 About Here]

Looking first to the effect of imagery on risk perceptions, the results were similar to what was found in the ANCOVA. Compared to the no-imagery condition, episodic imagery increased the perceived threat of flooding (unstandardized $b = .12$, $p = .014$) but did not

significantly influence the perceived threat of climate change ($b = .007, p = .907$); thematic imagery did not significantly influence the perceived threat of flooding ($b = -.03, p = .575$) or climate change ($b = .01, p = .800$). Compared to thematic imagery, episodic imagery significantly increased the perceived threat of flooding ($b = .15, p = .003$) but did not show a significant difference for the perceived threat of climate change ($b = -.008, p = .003$).

Party cues also showed a similar pattern to the ANCOVA; compared to the no-party cue condition, elite ($b = -.12, p = .013$) and public ($b = -.14, p = .005$) cues significantly lowered the perceived threat of flooding, but had no effect on the perceived threat of climate change (respectively, $b = -.08, p = .140$; $b = -.08, p = .171$). There was no difference between elite and public cues for the perceived threat of flooding ($b = .03, p = .470$) or climate change ($b = -.01, p = .835$).

Looking to downstream mediation, the perceived threat of flooding ($b = .12, p < .001$) and the perceived threat of climate change ($b = .43, p < .001$) both had a positive association with perceived collective efficacy, in support of H4a and H4b. Addressing RQ4, perceived collective efficacy had a positive association ($b = .42, p < .001$) with political action intentions.

Looking to indirect effects of imagery, compared to the no-imagery condition, the episodic imagery condition had significant positive indirect effects on political action intentions through the pathways of imagery $\rightarrow$ perceived threat of flooding $\rightarrow$ political action intentions ($b = .009, [LLCI = .001, ULCI = .020]$) and imagery $\rightarrow$ perceived threat of flooding $\rightarrow$ collective efficacy $\rightarrow$ political action intentions ($b = .006, [LLCI = .001, ULCI = .012]$). Compared to the thematic-imagery condition, the episodic imagery condition had significant positive indirect effects on political action intentions through the same respective pathways ($b = .01, [LLCI =$

.005, ULCI = .025]; $b = .009$, [LLCI = .004, ULCI = .015]). No other indirect effects from the imagery message factors were significant.

Looking to indirect effects of party cues, compared to the no-party cue condition, the elite party cue condition had significant negative indirect effects on political action intentions through the pathways of party cues → perceived threat of flooding → political action intentions ($b = -.01$, [LLCI = -.02, ULCI = -.001]) and party cues → perceived threat of flooding → collective efficacy → political action intentions ($b = -.006$, [LLCI = -.01, ULCI = -.001]). The public party cue message had similar negative effects through the same indirect pathways ($b = -.01$, [LLCI = -.02, -.002]; $b = \text{ULCI} = -.007$, [LLCI = -.01, ULCI = -.002]). No other indirect effects from the party cue factors were significant.

In addition to comparing the message conditions against each other, we also investigated how all message conditions compared to the no-message control condition. In order to improve readability, we provide full details of this analysis in the Appendix (Tables A1, A2, A3, and A4). To briefly summarize the OLS regression results comparing messages to the no-message control, messages only had a significant total effect on the perceived threat of flooding. All three imagery and party cue conditions interacted with political ideology on the perceived threat of flooding (Episodic imagery: $b = -.17$, $p < .001$; Thematic imagery: $b = -.15$, $p < .001$; No imagery: $b = -.11$, $p = .004$; Elite party cues: $b = -.15$, $p < .001$; Public party cues: $b = -.16$, $p < .001$; No party cues: $b = -.12$, $p = .002$). A Johnson-Neyman analysis revealed that all of the interactions were such that the messages, compared to the no-message control, increased the perceived threat of flooding for participants in the liberal and moderate range of political ideology, but did not have a significant effect for conservatives. No other interactions were

significant. The interactions were such that, for liberals and moderates, the indirect effects were very similar to those reported above comparing the message conditions, whereas there were no significant indirect effects for conservatives. See Figures 2 and 3 for a visual representation of the interactions. A comparison of the means of the perceived threat of flooding for the experimental factors at different levels of political ideology can be found in Table 2. Full details of the moderated-mediation model are provided in the Appendix.

[Insert Figure 2 About Here]

[Insert Figure 3 About Here]

[Insert Table 2 About Here]

Discussion

This study examined how party cues and imagery in news stories about the link between flooding and climate change may influence risk perceptions of flooding and climate change, collective efficacy, and intended political action. Table 3 provides a summary of all findings. Overall, we found that the inclusion of episodic impact imagery increased the perceived risk of flooding (in support of H1a) but not climate change (failing to support H1b), whereas thematic imagery did not have a significant influence on the perceived risk of flooding or climate change (failing to support H1a and H1b). When directly compared, episodic imagery increased risk perceptions of flooding compared to thematic imagery, but there was no difference between episodic and thematic imagery on risk perceptions of climate change (RQ1). The presence of party cues lowered the perceived threat of flooding (in support of H2a) but did not affect the perceived risk of climate change (failing to support H2b). There was no difference between elite and lay party cues in influencing risk perceptions (RQ2). Compared to news stories that did not

include imagery or party cues, political ideology did not moderate the influence of imagery or party cues (RQ3). While we did not find a significant interaction between political ideology and imagery when comparing between message conditions, it is important to note that Table 2 suggests that the use of imagery did not increase risk perceptions for conservatives. In addition, when comparing message conditions to the no-message control, there was a significant interaction such that conservatives were not influenced by the messages, but liberals and moderates were. The perceived risk of flooding (H4a) and climate change (H4b) both had a positive association with perceived collective efficacy, which aligns with motivated control predictions. Perceived collective efficacy, in turn, had a positive association with intended political action (RQ4). In combination, the presence of party cues and episodic imagery had a significant indirect effect on intended action through risk perceptions of flooding and collective efficacy.

[Insert Table 3 About Here]

While the indirect effects were significant, which provides important information about processes between theoretical constructs, there was not a significant total effect of party cues or imagery on collective efficacy or intentions for political action. Climate change messages typically have very small effect sizes (Rode et al., 2021), and that was the case in the present study as well. A η_p^2 of .01 is generally considered to be a small effect, so the observed η_p^2 of 0.004 for the influence of party cues and imagery on the perceived risk of flooding is a significant, but very small effect. These small effect sizes may help explain why we didn't observe total effects on more distal variables like perceived collective efficacy and intentions to engage in political behavior.

Overall, these findings make several important contributions to the literature. First, when comparing between message conditions, the lack of interactions between imagery, party cues, and political ideology indicates that imagery did not attenuate politicization on climate change as found in previous research (Bolsen et al., 2019a). The political divide between liberals and conservatives was not reduced by the presence of imagery, regardless of whether party cues were present. Thus, results of the present study suggest that the use of imagery in climate change messages is unlikely to reduce politicized responses on the issue. Furthermore, Table 2 suggests that news articles about climate change in general, or articles that include climate change imagery, are unlikely to increase risk perceptions for conservatives.

Second, imagery and party cues had a significant influence on the perceived risk of flooding but not the perceived risk of climate change. This is likely due to several factors - climate change opinions are likely more crystalized than flooding opinions; climate change operates in a polluted information environment, and climate change beliefs are strongly tied to social identity (Funk & Hefferon, 2019; Kahan, 2012; McCright et al., 2016; McCright & Dunlap, 2011; Nisbet, 2009). Overall, this finding shows that individuals may be sensitive to concrete threats depicted in climate change messages, such as flooding, even if these depictions do not alter risk perceptions of climate change as a whole. Climate advocates may be able to leverage this finding by utilizing narratives that focus on how taking action can address direct threats such as flooding or air pollution, rather than focusing on climate change as a whole. Previous research has shown this to be an effective communication strategy (Hart & Feldman, 2018, 2021).

Third, the failure to find a difference between the influence of public and elite party cues suggests that party divisions are important, but the source of these divisions may be less so. Overall, the results align with those of Linde (2020), in which information about conflicting party cues led to lower levels of risk perceptions, and the present study demonstrates that this finding is robust to whether the party cues come from elites or the lay public. This finding may have important implications for advocacy efforts that stress the difference between Democratic and Republican beliefs as a way to promote climate action. For example, an ad that promotes climate mitigation efforts undertaken by Democrats while attacking Republicans for failing to take sufficient action may have the unintended consequence of lowering risk perceptions. This finding also has implications for journalism, as it suggests that journalists' tendency to highlight partisan conflict and division in their reporting on climate change (Feldman et al., 2017; Hart & Feldman, 2014) can undermine public concern about the issue.

Looking to imagery, episodic imagery significantly influenced risk perceptions of flooding while thematic imagery did not compared to the no image conditions. This result aligns with research suggesting episodic imagery may be particularly effective at raising public concern about an issue (Rickard et al., 2017; Slovic, 2007). We also note that neither imagery condition had a negative influence on efficacy. This is consistent with some previous research (Hart & Feldman, 2016), and contrasts with literature that previously found that impact imagery can lower efficacy perceptions (O'Neill & Nicholson-Cole, 2009). Overall, the findings here suggest that impact imagery, and especially episodic imagery, can increase risk perceptions without lowering efficacy.

Looking to downstream factors, risk perceptions had a positive association with perceived collective efficacy, which aligns with predictions from some motivated control literature, and collective efficacy, in turn, had a positive association with intended political action. In past literature, there has been mixed support on the robustness of the positive relationship between collective efficacy and predispositions for climate action (Doherty & Webler, 2016; van Zomeren et al., 2008). The positive association we find adds strength to the proposition that individuals who feel a greater sense of collective efficacy are more likely to engage in political action. Taken together, these findings suggest that increasing the perception that climate change is a threat is unlikely to diminish the perceived efficacy of taking action, and potentially can help increase political action intentions.

Advantages of this study include a robust sample size with a diverse national sample and the use of carefully controlled experimental stimuli. However, as with any study, there are important limitations to consider. First, while stimulus sampling of imagery was utilized to prevent case-category confounds, it is still possible that there may be unique characteristics of the images used in the study that may have affected participant responses. It is also possible that if different language was used for the text of the experimental stimuli, participants may have had different responses. The influence of the experimental conditions provides clear causal pathways, but the use of cross-sectional data means that even though the ordering of variables in the mediation model was based on prior literature, the causal ordering of the mediators and ultimate dependent variable cannot be definitively established by the dataset. For collective efficacy and intentions for political action, we focused on climate change related perceptions and actions in order to better gauge how news stories may mobilize individuals

around climate change; however, we did not ask about efficacy and political action focused on flooding directly. Given that we only saw significant total effects on risk perceptions around flooding (but not risk perceptions of climate change), it is possible that if we had asked about collective efficacy and political action related to flooding we could have seen total effects there as well – it would be beneficial for future research to examine this question. Our variables measuring the perceived threat of flooding and climate change used single-item measures. While previous studies (Bolsen et al., 2019b; Kahan et al., 2015; Skurka et al., 2018) and leading survey organizations such as Gallup, PEW, and the European Social Survey have often used single item measures for the perceived threat of issues like climate change, it is important to note that single-item measures also have inherent limitations, such as the inability to calculate a reliability for the measurement. The use of the intended political action scale has been well established in the field, but it is important to keep in mind that political behavior is not directly measured, and intentions do not always lead to actual behavioral responses. Also, the results are limited to the U.S. context and may have limited generalizability to other parts of the world. Finally, this study focused on the flooding risk from climate change, but it is possible that results may have differed if a different climate risk, such as the increasing risk of heat stroke, or another type of disaster or extreme weather event, like wildfires, was the primary focus.

Overall, the results demonstrate that vivid, episodic imagery of floods, but not thematic, map imagery, can increase individuals' perceptions of flooding risk, whereas the presence of partisan political cues dampens flooding risk perceptions. While imagery and party cues have significant indirect effects on collective efficacy and political action, the lack of total effects on these variables indicates an overall limited effect on political engagement. The findings also

reveal the limits of using message features to shift climate change (as opposed to flooding) risk perceptions, as well as the challenges of reaching conservatives with climate change messages. Still, the effects on perceived flooding risks – and the role that risk perceptions play in driving collective efficacy and intentions – are important theoretically and practically. Of note, our results provide further support for motivated control processes in the context of climate change, given positive associations between perceived risk and efficacy, and between perceived efficacy and behavioral intentions. These results illuminate the indirect effects that imagery and party cues can have in driving motivated control processes and spurring climate action. Collectively, these findings demonstrate the need to account for imagery and the presence of party cues in future research and communication practice.

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Table 1

OLS Regression Results Predicting Risk Perceptions, Collective Efficacy, and Intentions to take Political Action

	Perceived Risk of Flooding	Perceived Risk of Climate Change	Perceived Collective Efficacy	Intentions to Take Political Action
	B (SE)	B (SE)	B (SE)	B (SE)
<i>Control Variables</i>				
Age	-0.0006 (.001)	-0.005 (.001)***	-0.004 (.001)***	-0.008 (.001)***
Gender (Female)	0.12 (.04)	0.18 (.05)***	0.17 (.03)***	-0.23 (.04)***
Political Ideology	-0.20 (.01)	-0.32 (.01)***	-0.06 (.01)***	-0.10 (.01)***
<i>Experimental Conditions</i>				
Episodic Imagery ^A	0.12 (.05)*	0.007 (.06)	-0.02 (.04)	-0.02 (.04)
Thematic Imagery ^A	-0.03 (.05)	0.01 (.06)	0.03 (.04)	-0.01 (.04)
Elite Party Cues ^B	-0.12 (.05)*	-0.08 (.06)	0.03 (.04)	-0.02 (.04)
Public Party Cues ^B	-0.14 (.05)**	-0.08 (.06)	-0.01 (.04)	0.01 (.04)
<i>Mediators</i>				
Perceived Risk of Flooding	-	-	0.12 (.02)***	0.08 (.02)***
Perceived Risk of Climate Change	-	-	0.44 (.02)***	0.26 (.02)***
Perceived Collective Efficacy	-	-	-	0.42 (.02)***
Constant	4.37 (.11)***	4.92 (.12)***	2.04 (.11)***	1.26 (.13)***
R ²	0.12	0.22	0.47	0.52

Note: Unstandardized regression coefficients are reported. *N* = 2,464 after listwise deletion across all variables.

^A No image message condition is the reference category

^B No party cues message condition is the reference category

*** $p < .01$, ** $p < .01$, * $p < .05$.

Table 2

Mean scores for Perceived Threat of Flooding, compared by Experimental Factors and Political Ideology.

Imagery Experimental Factor by Political Ideology			
	Liberal	Moderate	Conservative
	M (SD)	M (SD)	M (SD)
No-Message Control	3.67 (1.04)	3.46 (0.99)	3.43 (1.04)
No-Image Message	4.03 (0.97)	3.64 (1.03)	3.29 (1.17)
Thematic Image	4.03 (0.89)	3.71 (1.00)	3.15 (1.12)
Episodic Image	4.22 (0.88)	3.81 (1.01)	3.28 (1.08)

Party Cues Experimental Factor by Political Ideology			
	Liberal	Moderate	Conservative
	M (SD)	M (SD)	M (SD)
No-Message Control	3.67 (1.04)	3.46 (0.99)	3.43 (1.04)
No-Party Cues Message	4.12 (0.91)	3.77 (0.97)	3.39 (1.15)
Elite Party Cues	4.07 (0.95)	3.73 (1.05)	3.12 (1.07)
Public Party Cues	4.07 (0.90)	3.65 (1.03)	3.20 (1.14)

Note. While the continuous 7-point scale of political ideology was used for all interaction analyses, for ease of interpretation, the table above shows the grouped mean scores for liberals (1-3 on the 7-point political ideology scale), moderates (4 on the political ideology scale), and conservatives (5-7 on the political ideology scale).

Table 3*Summary of Findings*

Hypothesis / Research Question		Outcome
H1a	A climate change news story that includes flood imagery will increase the perceived threat of flooding relative to a news story without imagery.	- Episodic Imagery: Supported. - Thematic Imagery: Not supported.
H1b	A climate change news story that includes flood imagery will increase the perceived threat of climate change relative to a news story without imagery.	Not Supported
RQ1	Will vivid, episodic impact imagery differentially influence the perceived threat of flooding and climate change compared to thematic imagery?	- Perceived Flooding Threat: Episodic imagery led to higher risk perceptions compared to thematic imagery. - Perceived Climate Change Threat: No difference.
H2a	A climate change news story that includes party cues highlighting the political conflict around climate change will decrease the perceived threat of flooding relative to a news story without party cues.	Supported for both elite and lay party cues
H2b	A climate change news story that includes party cues highlighting the political conflict around climate change will decrease the perceived threat of climate change relative to a news story without party cues.	Not Supported
RQ2	Will elite party cues (Congressional representatives) differentially influence the perceived threat of flooding and climate change compared to party cues from lay partisans regarding the issue?	No
H3a	The presence of party cues and impact imagery in climate change news stories will interact to influence risk perceptions of flooding.	Not Supported

H3b	The presence of party cues and impact imagery in climate change news stories will interact to influence risk perceptions of climate change.	Not Supported
RQ3	Will political ideology moderate the influence of party cues and impact imagery on public perceptions of climate change and flooding?	- Perceived Flooding Threat: No significant interactions when comparing between messages. Significant interactions when comparing against control - all message conditions increase risk perceptions for liberals and moderates, but have no effect for conservatives. - Perceived Climate Change Threat: No significant interactions, whether comparing between messages or against control.
H4a	The perceived risk of flooding will have a positive association with perceived collective efficacy to address climate change.	Supported
H4a	The perceived risk of climate change will have a positive association with perceived collective efficacy to address climate change.	Supported
RQ4	Will perceived collective efficacy have a positive association with intentions to take political action?	Yes
Additional Findings of Note.		Message conditions only had significant total effects on the perceived risk of flooding. Message conditions had significant indirect effects on intentions for political action through the perceived risk of flooding and perceived collective efficacy, with all tables for mediation analyses provided in the appendix.

Figure 1

Mediation Model for Testing

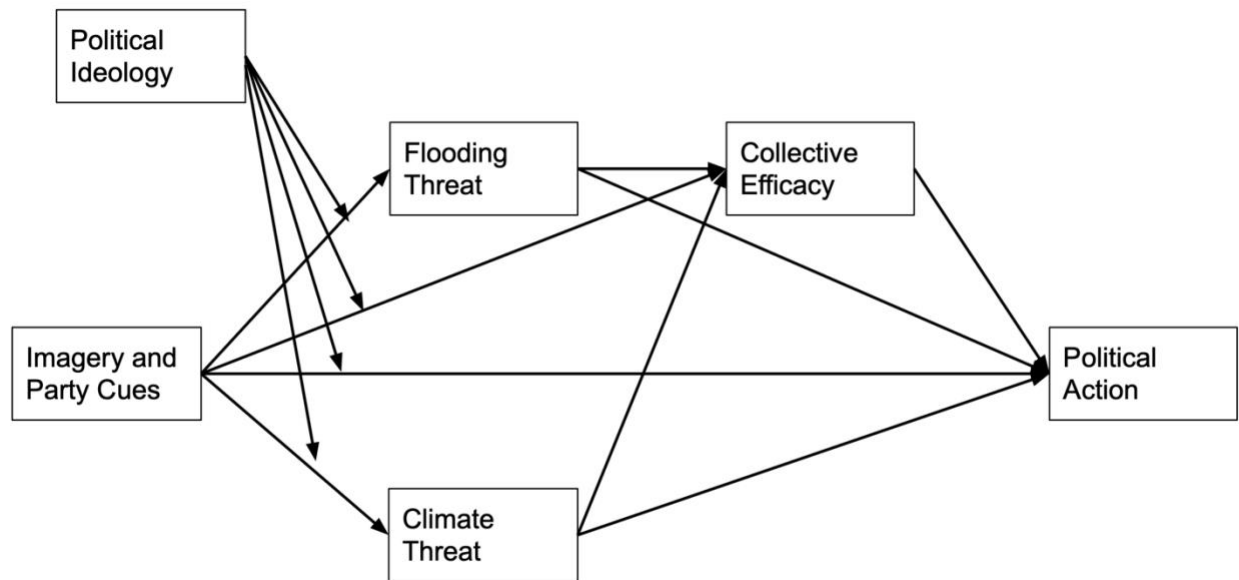


Figure 2

Imagery Conditions by Political Ideology on Perceived Flood Risk

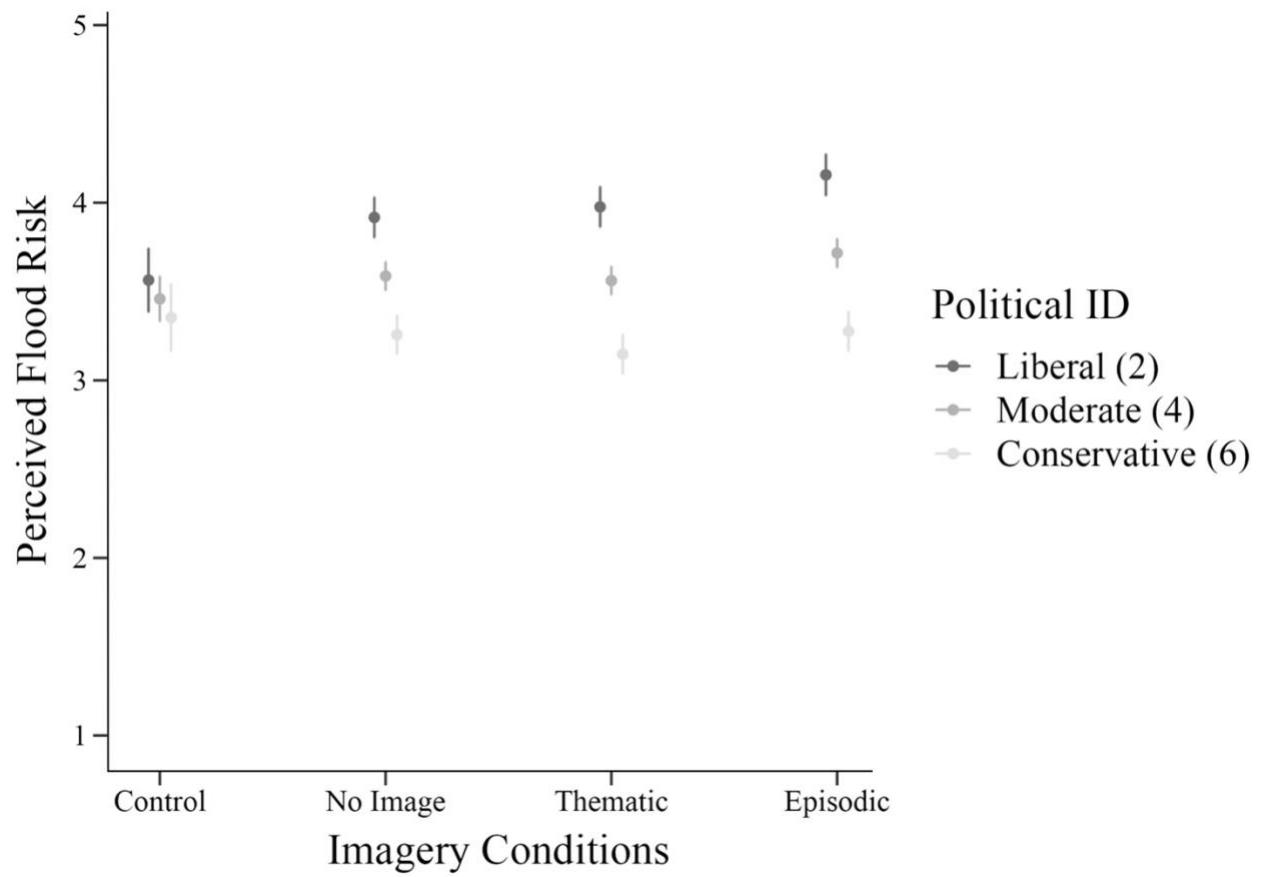
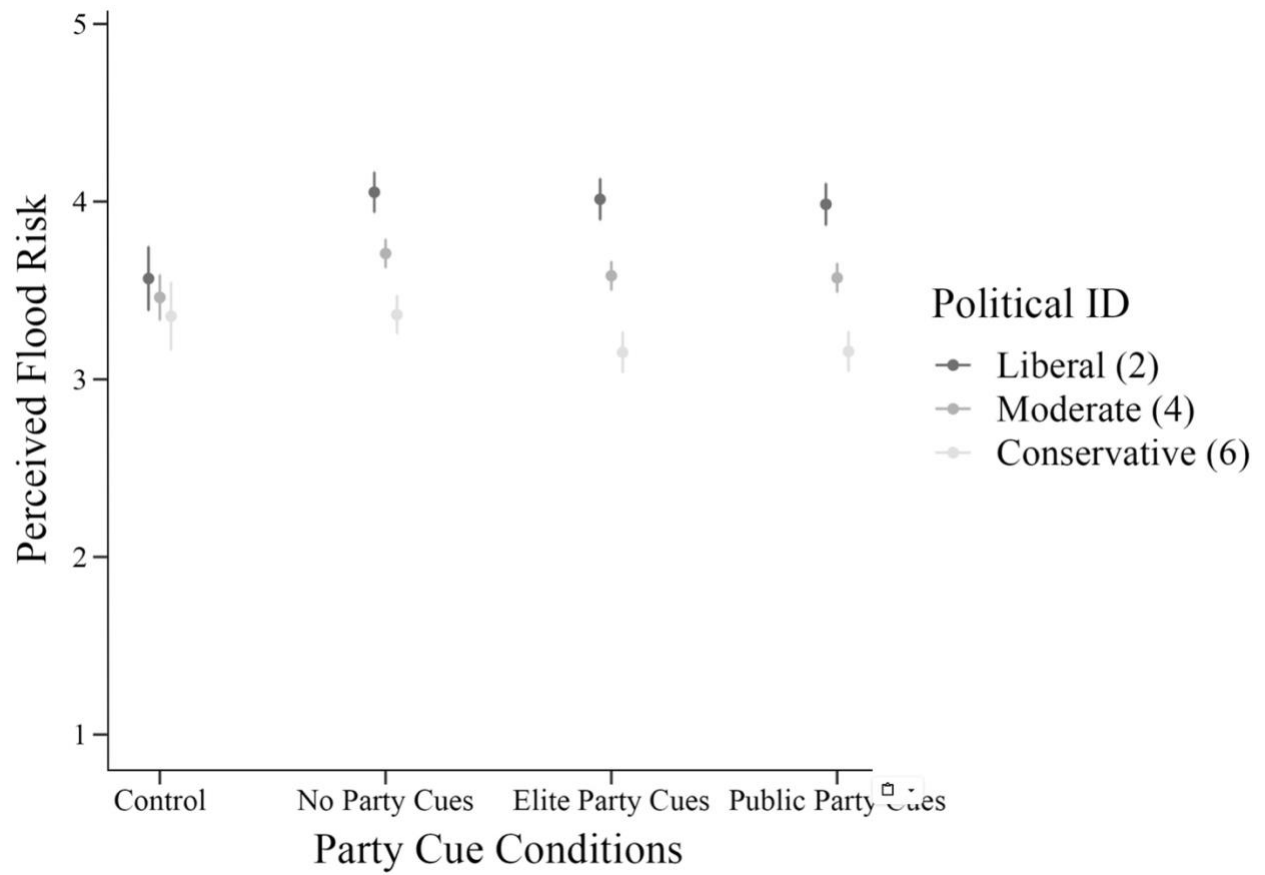


Figure 3

Party Cue Conditions by Political Ideology on Perceived Flood Risk



Appendix

The following tables show the OLS regression output for imagery and party cues against the no-message control. The analysis only looks at one factor at a time, as including both factors in the model at the same creates a singularity. In the OLS regressions presented, non-significant interactions have been culled, as this is what was used for the indirect effect analysis in the moderated-mediation model. Tables A1 and A2 show the OLS regression results and indirect effects for the moderated-mediation model of the imagery conditions against control, and tables A3 and A4 show the output for party cues against the no message control.

Table A1

Mediation Model for Imagery Against No Message Control

	Perceived Risk of Flooding	Perceived Risk of Climate Change	Perceived Collective Efficacy	Intentions to Take Political Action
	B (SE)	B (SE)	B (SE)	B (SE)
<i>Control Variables</i>				
Age	-0.001 (.001)	-0.005 (0.001)***	-0.004 (0.0009)***	-0.008 (0.001)***
Gender (Female)	0.11 (0.04)**	0.18 (0.04)***	0.15 (0.03)***	-0.22 (0.03)***
Political Ideology	-0.05 (0.03)	-0.32 (0.01)***	-0.06 (0.009)***	-0.10 (0.01)***
<i>Experimental Conditions</i>				
Episodic Imagery	0.93 (0.17)***A	0.01 (0.08)	-0.05 (0.06)	-0.03 (0.06)
Thematic Imagery	0.72 (0.17)***A	0.02 (0.08)	-0.002 (0.06)	-0.02 (0.06)
No Imagery	0.58 (0.17)***A	0.003 (0.08)	-0.03 (0.06)	-0.002 (0.06)
<i>Interactions</i>				
Episodic Imagery *				
Political Ideology	-0.17 (0.04)***B	-	-	-

Thematic Imagery *				
Political Ideology	-0.15 (0.04)*** ^C	-	-	-
No Imagery *				
Political Ideology	-0.11 (0.04)** ^D	-	-	-
<i>Mediators</i>				
Perceived Risk of Flooding	-	-	0.11 (0.02)***	0.07 (0.02)***
Perceived Risk of Climate Change	-	-	0.44 (0.02)***	0.27 (0.02)***
Perceived Collective Efficacy	-	-	-	0.40 (0.02)***
Constant	3.61 (0.17)***	4.86 (0.12)***	2.05 (0.11)***	1.26 (0.13)***
R ²	0.11	0.22	0.46	0.52

Note: Unstandardized regression coefficients are reported. *N* = 2,736 after listwise deletion across all variables.

Non-significant interactions have been removed from the regression equations.

^A no-message control condition is the reference category

Johnson-Neyman points of significance ranges for political ideology: ^B [1.00 - 4.64], ^C [1.00 - 3.76], ^D [1.00 - 3.90]

*** *p* < .01, ** *p* < .01, * *p* < .05.

Table A2*Indirect Effects of Imagery Message Factors and Different Levels of Political Ideology*

Indirect Effect of Episodic Imagery: Imagery --> Risk of Flooding --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.04	[0.01 - 0.07]
4 (Moderate)	0.02	[0.005 - 0.04]
6 (Conservative)	-0.005	[-0.02 - 0.01]
Indirect Effect of Episodic Imagery: Imagery --> Risk of Flooding --> Collective Efficacy --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.03	[0.01 - 0.04]
4 (Moderate)	0.01	[0.005 - 0.02]
6 (Conservative)	-0.004	[-0.01 - 0.007]
Indirect Effect of Thematic Imagery: Imagery --> Risk of Flooding --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.03	[0.009 - 0.05]
4 (Moderate)	0.007	[-0.002 - 0.02]
6 (Conservative)	-0.01	[-0.04 - 0.002]
Indirect Effect of Thematic Imagery: Imagery --> Risk of Flooding --> Collective Efficacy --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.02	[0.009 - 0.03]
4 (Moderate)	0.005	[-0.002 - 0.01]
6 (Conservative)	-0.009	[-0.02 - 0.001]
Indirect Effect of No Imagery: Imagery --> Risk of Flooding --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.02	[0.007 - 0.05]
4 (Moderate)	0.009	[-0.001 - 0.02]

6 (Conservative)	-0.007	[-0.03 - 0.009]
Indirect Effect of No Imagery: Imagery --> Risk of Flooding --> Collective Efficacy --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.02	[0.006 - 0.03]
4 (Moderate)	0.006	[-0.0007 - 0.01]
6 (Conservative)	-0.004	[-0.02 - 0.006]

Table A3*Mediation Model for Party Cues Against No Message Control*

	Perceived Risk of Flooding	Perceived Risk of Climate Change	Perceived Collective Efficacy	Intentions to Take Political Action
	B (SE)	B (SE)	B (SE)	B (SE)
<i>Control Variables</i>				
Age	-0.0008 (0.001)	-0.005 (0.001)***	-0.004 (0.0009)***	-0.008 (0.001)***
Gender (Female)	0.11 (0.04)	0.17 (0.04)***	0.15 (0.03)***	-0.22 (0.03)***
Political Ideology	-0.05 (0.03)	-0.33 (0.1)***	-0.06 (0.01)***	-0.10 (0.01)***
<i>Experimental Conditions</i>				
Elite Party Cues	0.77 (0.17)***A	-0.02 (0.08)	-0.004 (0.06)	-0.03 (0.06)
Public Party Cues	0.73 (0.17)***A	-0.01 (0.08)	-0.04 (0.06)	-0.002 (0.06)
No Party Cues	0.72 (0.17)***A	0.06 (0.08)	-0.03 (0.06)	-0.01 (0.06)
<i>Interactions</i>				
Elite Party Cues * Political IdeologyA	-0.16 (0.04)***B	-	-	-
Public Party Cues * Political IdeologyA	-0.15 (0.04)***C	-	-	-
No Party Cues * Political IdeologyA	-0.12 (0.04)**D	-	-	-
<i>Mediators</i>				
Perceived Risk of Flooding	-	-	0.11 (0.02)***	0.07 (0.02)***
Perceived Risk of Climate Change	-	-	0.44 (0.02)***	0.27 (0.02)***
Perceived Collective Efficacy	-	-	-	0.40 (0.02)***

Constant	3.61 (0.17)***	4.87 (0.12)***	2.06 (0.11)***	1.26 (0.13)***
R ²	0.11	0.22	0.46	0.52

Note: Unstandardized regression coefficients are reported. *N* = 2,736 after listwise deletion across all variables.

Non-significant interactions have been removed from the regression equations.

^A no-message control condition is the reference category

Johnson-Neyman points of significance ranges for political ideology: ^B [1.00 - 3.88], ^C [1.00 - 3.80], ^D [1.00 - 4.78]

*** *p* < .01, ** *p* < .01, * *p* < .05.

Table A4*Indirect Effects of Party Cue Message Factors and Different Levels of Political Ideology*

Indirect Effect of Elite Party Cues: Party Cues --> Risk of Flooding --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.03	[0.01 - 0.06]
4 (Moderate)	0.008	[-.001 - 0.02]
6 (Conservative)	-0.01	[-0.04 - 0.001]
Indirect Effect of Elite Party Cues: Party Cues --> Risk of Flooding --> Collective Efficacy --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.02	[0.01 - 0.03]
4 (Moderate)	0.006	[-0.0009 - 0.01]
6 (Conservative)	-0.009	[-0.02 - 0.001]
Indirect Effect of Public Party Cues: Party Cues --> Risk of Flooding --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.03	[0.01 - 0.05]
4 (Moderate)	0.008	[-0.002 - 0.02]
6 (Conservative)	-0.01	[-0.04 - 0.003]
Indirect Effect of Public Party Cues: Party Cues --> Risk of Flooding --> Collective Efficacy --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.19	[0.009 - 0.03]
4 (Moderate)	0.005	[-0.001 - 0.01]
6 (Conservative)	-0.01	[-0.02 - 0.002]
Indirect Effect of No Party Cues: Party Cues --> Risk of Flooding --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.03	[0.01 - 0.06]
4 (Moderate)	0.02	[0.05 - 0.03]

6 (Conservative)	0.0006	[-0.01 - 0.02]
Indirect Effect of No Party Cues: Party Cues --> Risk of Flooding --> Collective Efficacy --> Intentions for Political Action		
Level of Political Ideology	Point Estimate	95% Confidence Interval
2 (Liberal)	0.02	[0.01 - 0.04]
4 (Moderate)	0.01	[0.004 - 0.02]
6 (Conservative)	0.0004	[-0.01 - 0.01]

Figure A1

Example Stimulus: Thematic Imagery and Elite Party Cues

AP

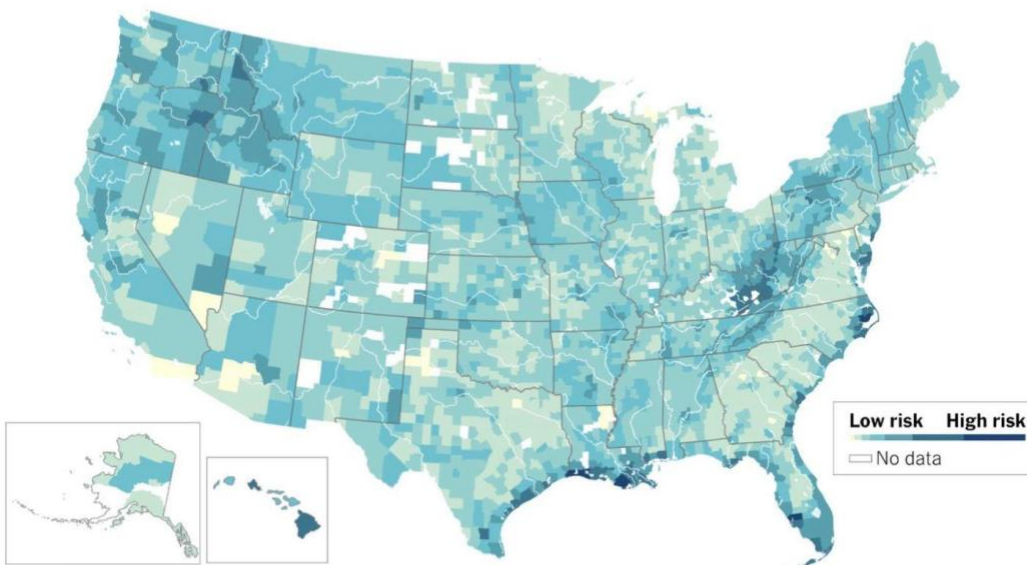
October 11, 2021 10:30 AM EDT

New US government report indicates climate change likely to increase risk of flooding, politicians divided on the issue

ASSOCIATED PRESS



Flood Risk in the United States



Climate change is increasing the risk of floods in the United States. *AP / Staff*

WASHINGTON (AP) – A report released last month by the Federal Emergency Management Agency (FEMA) indicates that climate change is increasing the risk of flooding throughout the United States. Rising sea levels and heavier rainstorms, driven by climate change, are creating longer and more intense wet seasons that make residential neighborhoods susceptible to flooding. As these floods become more frequent and severe, the report finds, the number of homes at risk for damage or destruction from these events will continue to grow. In the past decade alone, deadly flooding has caused billions of dollars in damages and forced hundreds of thousands to flee their homes.

Politicians, however, remain divided on the link between climate change and flooding. A majority of Congressional Democrats have stated that climate change is increasing the frequency and severity of flooding, while a majority of Congressional Republicans remain skeptical.

Figure

Figure A2

Example Stimulus: Episodic Imagery and Public Party Cues

AP

October 11, 2021 10:30 AM EDT

New US government report indicates climate change likely to increase risk of flooding, public divided on the issue

ASSOCIATED PRESS



Climate change is increasing the risk of floods in the United States. *AP / Staff*

WASHINGTON (AP) – A report released last month by the Federal Emergency Management Agency (FEMA) indicates that climate change is increasing the risk of flooding throughout the United States. Rising sea levels and heavier rainstorms, driven by climate change, are creating longer and more intense wet seasons that make residential neighborhoods susceptible to flooding. As these floods become more frequent and severe, the report finds, the number of homes at risk for damage or destruction from these events will continue to grow. In the past decade alone, deadly flooding has caused billions of dollars in damages and forced hundreds of thousands to flee their homes.

The public, however, remains divided on the link between climate change and flooding. Recent polling from the Pew Research Center finds that a majority of Democrats are convinced that climate change is increasing the frequency and severity of flooding, while a majority of Republicans remain skeptical.